

Verification Packet

SOXL/TQQQ Rotation with cash daily scanner

Purpose: document the scanner trading rules and the available in-sample/out-of-sample performance evidence so a third-party model can verify the setup and numbers.

Current run/status artifact

Field	Value
Generated	2026-05-29T18:58:24-04:00
Mode	stale_fallback
Can trade live	False
Executed trade	False
Status reason	alpaca_positions_unavailable: Alpaca request failed (GET /v2/positions/SOXL): <urlopen..

Primary source files

- Rules: scripts/soxl_tqqq_cash_signal_scanner.py and scripts/soxl_tqqq_cash_signal_scanner_stdlib.py.
- IS/OOS selection table: SOXL_TQQQ_InSample_Top10_OutSample_Rank.csv.
- Latest OOS metrics: reports/soxl_tqqq_rotation_cash_oos_performance_summary_metrics.csv.
- Annual OOS return/drawdown: reports/soxl_tqqq_rotation_cash_oos_performance_annual_return_drawdown.csv.
- Operational status: reports/soxl_tqqq_cash_run_status.json.

Trading Rules

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Universe and data

- Tradable symbols: SOXL and TQQQ. Context trend symbol: QQQ.
- Scanner timeframe: daily bars, evaluated after the daily close.
- Price data: adjusted daily close history for SOXL, TQQQ, and QQQ.
- Target output is exactly one of SOXL, TQQQ, or CASH.

Base rotation

- Compute each ETF's relative momentum score as 63-trading-day return minus 0.5 times annualized realized volatility.
- Skip the most recent 10 trading days in both the return and volatility inputs.
- Initial raw choice is SOXL when SOXL score is greater than or equal to TQQQ score; otherwise TQQQ.
- Trend preference: if the selected ETF is below SMA50 and the alternate ETF is above SMA50, choose the alternate ETF.
- Hysteresis: switch the selected ETF only when the absolute score spread is at least 5 percentage points.
- Rebalance cadence: monthly. The chosen target is only refreshed on the first trading day of a new calendar month, then forward-filled.

Cash filter

- Cash SMA window: 150 trading days for the currently selected ETF and QQQ.
- Exit to CASH when both the selected ETF and QQQ close below their SMA150 levels.
- Re-enter risk-on when either the selected ETF or QQQ closes at or above SMA150 plus 1%.
- The selected ETF itself has no exit buffer; re-entry buffer is 1%.

Execution logic

- If target is CASH, sell any open SOXL/TQQQ positions.
- If target is SOXL or TQQQ, sell the non-target ETF first, then buy the target ETF if it is not already held.
- Default position size in the automation command is 1 share.
- Extended-hours execution uses Alpaca DAY limit orders with `extended_hours=True` and limit offset 0%.
- Default Alpaca mode is paper trading unless `--live` is supplied.
- Fallback mode: if live data or Alpaca is unavailable, write a `stale_fallback` status and skip trading.

IS/OOS Selection Audit

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Apples-to-apples comparison from SOXL_TQQQ_InSample_Top10_OutSample_Rank.csv

Strategy	IS period	IS return	IS DD	OOS period	OOS return	OOS DD	OOS Sharpe
Scanner cash strategy	2010-06-25 to 2019-12-20	2,523.18%	-52.16%	2020-01-02 to 2026-05-20	10,570.12%	-50.78%	1.42
SOXL buy-and-hold	2010-06-25 to 2019-12-20	3,337.59%	-72.61%	2020-01-02 to 2026-05-20	819.19%	-90.46%	0.87
TQQQ buy-and-hold	2010-06-25 to 2019-12-20	4,619.32%	-58.08%	2020-01-02 to 2026-05-20	605.45%	-81.66%	0.79

Interpretation

- The scanner cash strategy had lower IS drawdown than SOXL buy-and-hold and slightly lower IS drawdown than TQQQ buy-and-hold in this selection audit.
- The same selection-audit OOS slice shows substantially higher cumulative return and lower drawdown than both SOXL and TQQQ buy-and-hold.
- These values are selection-audit values through 2026-05-20; the latest scanner OOS refresh through 2026-05-22 is shown on the next page.

Latest OOS Refresh

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Strategy	OOS period	Return	CAGR	Max DD	Sharpe	Sortino	Calmar
Scanner cash strategy	2020-01-02 to 2026-05-22	9,826.00%	105.47%	-53.40%	1.403	1.71	1.975
SOXL buy-and-hold	2020-01-02 to 2026-05-22	911.32%	43.68%	-90.46%	0.888	1.282	0.483
TQQQ buy-and-hold	2020-01-02 to 2026-05-22	617.72%	36.16%	-81.66%	0.793	1.046	0.443

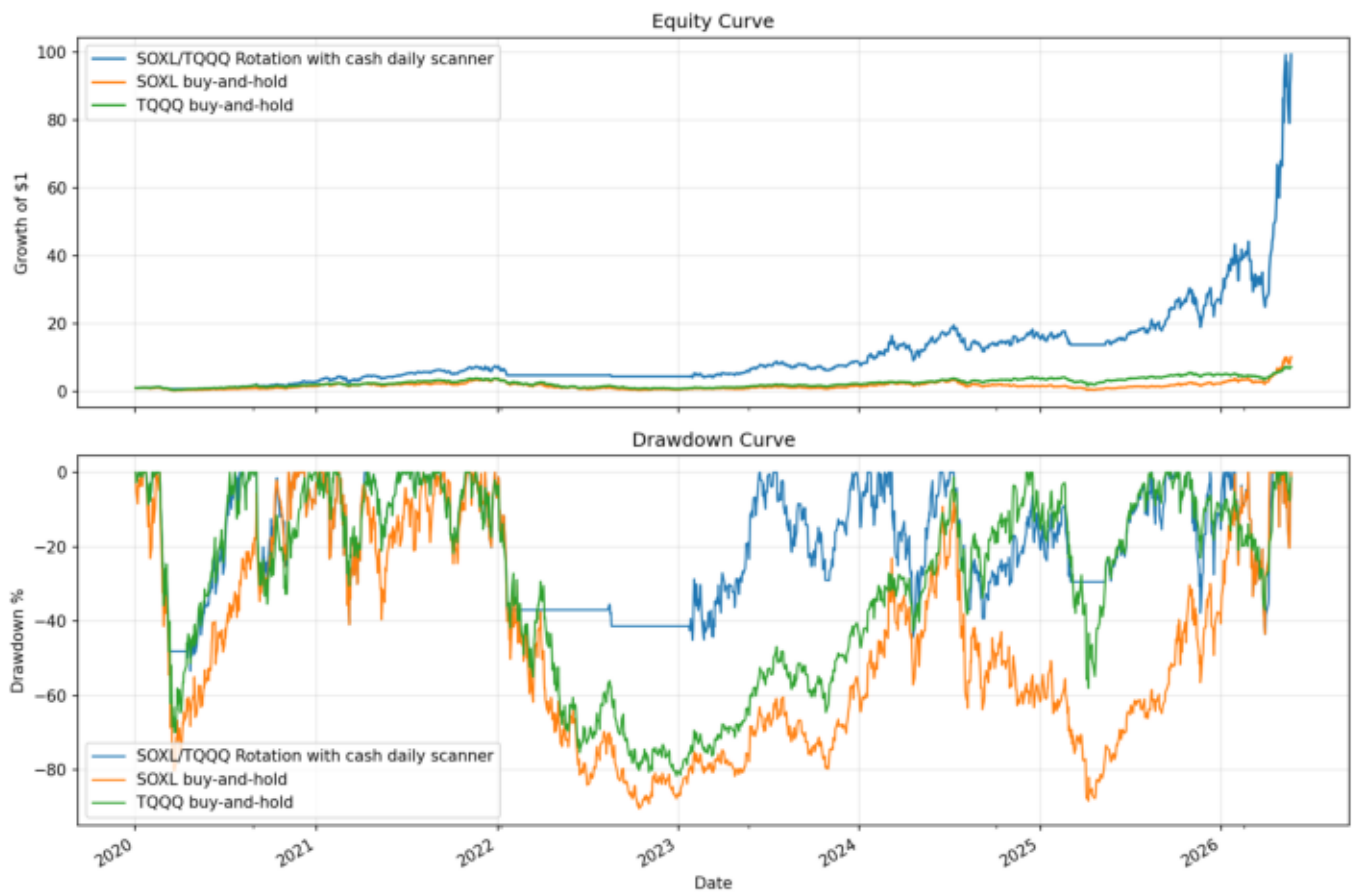
Annual OOS return / drawdown

Year	Scanner ret	Scanner DD	SOXL ret	SOXL DD	TQQQ ret	TQQQ DD
2020	193.26%	-53.40%	60.06%	-80.38%	100.13%	-69.92%
2021	140.98%	-40.92%	121.90%	-40.92%	91.35%	-30.43%
2022	-40.63%	-40.63%	-86.48%	-90.34%	-79.68%	-81.02%
2023	99.58%	-29.00%	237.81%	-48.99%	204.92%	-33.14%
2024	86.60%	-44.30%	-1.80%	-62.15%	66.72%	-37.37%
2025	67.57%	-37.78%	52.90%	-76.46%	35.21%	-56.84%
2026	242.01%	-43.47%	303.39%	-43.47%	48.94%	-33.49%

Latest OOS source period ends 2026-05-22. Annual 2026 values are year-to-date in the source file.

OOS Curves

SOXL/TQQQ Rotation with cash daily scanner



Source: reports/soxl_tqqq_rotation_cash_oos_performance_curves.png

Verification Notes

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Assumptions and caveats

- Backtest metrics are based on local generated artifacts; no new market data was downloaded for this packet.
- The strategy reports use adjusted daily prices and zero explicit trading cost unless stated otherwise.
- The strategy is a daily scanner whose base SOXL/TQQQ rotation updates monthly; those statements are not contradictory.
- The active runner uses a dependency-free scanner for operational robustness, but it implements the same rule set as the pandas/yfinance scanner.
- If run_status mode is stale_fallback, the scanner did not place trades during that run; it intentionally reused the prior signal and wrote the blocker reason.

Recommended Claude checks

- Confirm StrategyConfig values: 63 lookback, 10 skip, SMA50 trend, 5% hysteresis, SMA150 cash filter, 1% re-entry buffer.
- Confirm monthly rebalance is applied after daily score calculation and before cash filtering.
- Confirm execution path sells non-target exposure before buying the target, and does not trade in stale_fallback mode.
- Compare the IS/OOS table values against the listed CSV sources rather than mixing files with different OOS end dates.